Janardan Revuru

Solution Architecture and Delivery · Cloud-Native, Microservices, Event-Driven and AI-RAG Systems

Bengaluru, India 560100 | +91 8050780608 | janardan.revuru@gmail.com
LinkedIn: linkedin.com/in/rjanardan | Portfolio: janalogy.com | Research: knowledgesidekick.org

Professional Summary

Architected end-to-end cloud-native and AI solutions, translating business requirements into scalable, resilient designs across compute, containers, Kubernetes, data, messaging and integration. As a hands-on Director of Engineering, led management and supportability software across enterprise data-protection appliances and distributed storage systems, covering REST APIs, microservices, Kubernetes system management, PostgreSQL and observability. Designed API-first, modernization and multi-cloud solutions across AWS, Azure, GCP, S3, VMware and Alibaba. Specialist depth in grounded, non-hallucinating LLM systems: retrieval-augmented generation, vector search, knowledge representation and agent frameworks, built on doctoral research into agent-to-agent communication. Proven communicator who translates between technical and business stakeholders, runs discovery and design workshops, and leads architecture governance.

Architecture Practice

Solution design and enterprise architecture

- Translate business and functional requirements into scalable, resilient, cost-effective architectures spanning compute, containers, Kubernetes, data, messaging and integration.
- Design cloud-native microservices and API-first solutions; led API-first compliance across a product portfolio using OpenAPI and Redfish.
- Lead architecture assessments, fit-gap analysis, modernization and cloud-migration strategy, including the standard rehost, replatform, refactor, repurchase, retire and retain decision framework.
- Produce solution design documents, architecture decision records, and infrastructure-as-code blueprints; drive architecture governance with technology advisory roles at two universities.

AI and machine-learning architecture

- Design retrieval-augmented generation pipelines with vector databases (PostgreSQL with pgvector, and related stores), embedding-model selection, chunking strategy and latency and cost tradeoffs.
- Architect agentic workflows and multi-agent systems, with tool selection, memory and context management, guardrail placement and human-in-the-loop checkpoints; grounded in doctoral research on agent-to-agent communication.
- Design enterprise LLM integration patterns: prompt engineering as architecture, model serving tradeoffs (batching, quantization, serverless vs dedicated inference), and evaluation and observability for grounding, factual accuracy and hallucination control.
- Architect knowledge-layer systems for grounded, non-hallucinating answers using standards (RDF, OWL, SKOS, JSON-LD), provenance, and reverse traceability to source.

Cloud, data and integration

- Architect on-premises, hybrid-cloud and multi-cloud deployments; validated and deployed solutions on AWS, Azure, GCP, S3, VMware and Alibaba, with certified interoperability against a SAP customer and Google Cloud.
- Design event-driven and integration architectures with messaging and streaming, API management, and enterprise integration patterns.
- Model relational and vector data with PostgreSQL, pgvector, SQLAlchemy and Alembic; design OLTP and analytics data flows.

Selected Outcomes

API-first platform · led engineering-side of four version-1 enterprise products, one design system, API-first compliance (OpenAPI, Redfish), and 3 granted UX patents (co-inventor)

Kubernetes system management · architected system management of a distributed Kubernetes-based data-protection product with five early-preview customers

Grounded AI · built a knowledge layer for non-hallucinating answers: wiki creation in the agentic loop, reverse traceability, scenario analysis, 80% cost saving on system-test scenarios

Multi-cloud deployment · API-based lifecycle management for Data Domain Virtual Edition across AWS, Azure and GCP reaching 25 cloud deployments

Observability · integration with a SaaS monitoring platform (reaching 40% of systems) covering telemetry, alerting and continuous upgrades

Experience

Director of Engineering, Dell Technologies, Aug 2021 to Sep 2026

- Led a 60-engineer line (5 leaders) responsible for management and supportability software across PowerProtect DD and a distributed Kubernetes-based data-protection product; built the India function from 2 to 100.
- As AI champion of the DataDomain Appliances business unit, drove agentic-AI workflows across about 600 engineers with spec-driven development on an AI harness (3x productivity, 30% higher engagement), and worked with Dell IT to fine-tune code models to a million-line codebase.
- Drove automation-first delivery on an established CI/CD pipeline with production observability and field telemetry.

Senior Manager, Dell Technologies, Dec 2016 to Jul 2021

- Led API-based lifecycle management for Data Domain Virtual Edition across AWS, Azure and Google Cloud (25 cloud deployments); modernized a web stack from Google Web Toolkit to Angular; reduced support cases by 50% and mean time to resolve by 80%.

Engineering Manager, Hewlett Packard Enterprise, Mar 2011 to Dec 2016

- Managed three teams building management software for an HPE high-availability clustering product; supervised a multi-cluster manager architecture that cut customers' operational cost by 20%; oversaw the disaster-recovery program in partnership with EMC.

Skills

Kubernetes | Docker | AWS / Azure / GCP | REST APIs (OpenAPI, Redfish) | Microservices | Event-driven architecture | PostgreSQL / pgvector | SQLAlchemy / Alembic | Infrastructure as Code (Terraform) | CI/CD | Observability | GraphQL / gRPC | Model Context Protocol | RAG / vector databases | LangGraph / LangChain | PyTorch / TensorFlow / Hugging Face | RDF / OWL / JSON-LD | Architecture governance

Technologies I Am Building Now

Deliberate build over the coming quarter alongside the role: enterprise-integration and streaming patterns (Kafka, API gateway), deeper cloud-native application development, exposure to Databricks and Snowflake data platforms, and reinforcement of infrastructure-as-code with a major cloud provider. These are current building targets, stated transparently.

Kafka / event streaming | Databricks / Snowflake | Cloud-native application build (Terraform, foundational) | Streamlit / GenAI application UI

Education and Community

- **PhD in Artificial Intelligence**, REVA University (expected June 2028); **M.Tech Data Science**, BITS Pilani, 2024; **B.E. Computer Science**; **MBA Marketing**. 3 granted US patents (co-inventor).
- Founder of technology communities with 22K members and 100+ events across India; academic council (Madanapalle Institute of Technology and Science) and board of studies (Kishkindha University) roles.
- Authoring *Mastering TypeScript* for BPB Publications, reinforcing TypeScript and front-end architecture fluency for developer-facing platforms.